IGCSE 0607 Practice Questions — E2.5

Equations

Total Questions: 39 — **Total Marks:** 260 Marks

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1 Solve the equation.

$$\frac{3x-1}{4} = \frac{x+5}{3}$$

$x =$ ..... **[3]**

2 Solve the simultaneous equations.
You must show all your working.

$4x + 3y = 5$

$2x - 5y = 22$

$x =$ ..... , $y =$ ..... **[3]**

3 **(a)** Solve the equation.

$$4(x - 3) = 2x + 10$$

$x =$ ..... **[2]**

(b) Rearrange the formula to make x the subject.

$$y = \frac{2x - 1}{5}$$

$x =$ ..... **[2]**

4 **(a)** Solve the equation.

$3(2x - 1) = 5x + 8$

$x =$ ..... **[2]**

(b) Rearrange the formula to make x the subject.

$y = 4x - 7$

$x =$ ..... **[2]**

5 **(a)** Solve the equation.

$3(2x - 1) = 4x + 11$

$x =$ ..... **[2]**

(b) Solve the simultaneous equations.

$3x + 2y = 16$

$x - y = 2$

$x =$ ..... , $y =$ ..... **[3]**

6 **(a)** Solve

$$5x - 3 = 2(x + 6)$$

$x =$ ..... **[2]**

(b) Solve

$$\frac{x+2}{3} + \frac{2x-1}{4} = 3$$

$x =$ ..... **[3]**

7 **(a)** Solve the equation.

$$3(2x - 5) = 4x + 7$$

$x =$ ..... **[3]**

(b) Rearrange the formula to make x the subject.

$$y = \frac{2x + 3}{5}$$

$x =$ ..... **[2]**

8 **(a)** Solve the equation.

$$4(2x - 1) = 3x + 11$$

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$$y = \frac{3x - 5}{2}$$

$x =$ ..... **[3]**

9 Consider the following algebraic tasks.

(a) Solve the equation.

$$5(x-2) = 3x + 14$$

$x =$ ..... **[3]**

(b) Rearrange the formula to make b the subject.

$$A = \frac{h(a+b)}{2}$$

$b =$ ..... **[3]**

10 **(a)** Solve.

$4(x - 3) = 2x + 7$

$x =$ ..... **[3]**

(b) Solve the simultaneous equations.
You must show all your working.

$5x - 2y = 16$

$3x + 4y = 7$

$x =$ ..... , $y =$ ..... **[3]**

11 Rearrange the formula to make x the subject.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

12 **(a)** Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 12$

$5x - 3y = 1$

$x =$ ..... , $y =$ ..... **[3]**

(b) Solve the equation by factorisation.

$2x^2 - 5x - 12 = 0$

$x =$ ..... , $x =$ ..... **[3]**

13 **(a)** Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 12$

$5x - 3y = 1$

$x =$ ..... , $y =$ ..... **[4]**

(b) Solve the equation by factorisation.

$2x^2 - 7x - 15 = 0$

$x =$ ..... , or $x =$ ..... **[3]**

14 **(a)** Solve the simultaneous equations.
You must show all your working.

$3x - 2y = 19$

$4x + y = 18$

$x =$ ..... , $y =$ ..... **[3]**

(b) Rearrange the formula to make p the subject.

$$T = \frac{3p + 4}{p - 2}$$

$p =$ ..... **[4]**

15 **(a)** Solve the simultaneous equations.
You must show all your working.

$2x - y = 9$ $3x + 2y = 10$

$x =$ ..... , $y =$ ..... **[4]**

(b) Solve the equation.

$$\frac{2x}{3} - \frac{x-4}{4} = \frac{5}{2}$$

$x =$ ..... **[4]**

16 Solve the following equations.

(a) Solve the simultaneous equations.
You must show all your working.

$4x + 3y = 5$

$2x - 5y = 9$

$x =$ ..... , $y =$ ..... **[4]**

(b) Solve the equation by factorisation.

$3x^2 - 10x - 8 = 0$

$x =$ ..... or $x =$ **[4]**

17 Solve the equation.

$$\frac{2}{x-1} + \frac{1}{x-2} = 1$$

Give your answers in the form $a \pm \sqrt{b}$, where a and b are integers.
You must show all your working.

$x =$ ..... **[6]**

18 **(a)** Rearrange the formula to make x the subject.

$$p = \frac{3x - 1}{x + 4}$$

$x =$ ..... **[4]**

(b) Solve the equation by factorising.

$$2x^2 - 5x - 12 = 0$$

$x =$ ..... , $x =$ ..... **[3]**

19 **(a)** A rectangle has length $(3x + 2)$ cm and width $(x - 1)$ cm.
The perimeter of the rectangle is 42 cm.

Work out the value of x.

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$$p = \sqrt{\frac{x-3}{x}}$$

$x =$ ..... **[4]**

20 **(a)** Make x the subject of the formula.

$$y = \frac{2x + 5}{x - 3}$$

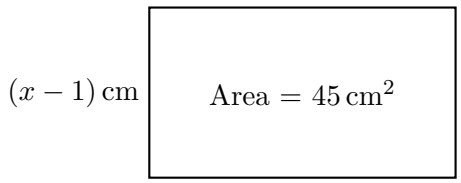
$x =$ ..... **[3]**

(b) Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{x + 1}{3} - \frac{2}{x - 2} = 1$$

$x =$ ..... , $x =$ ..... **[6]**

21 The diagram shows a rectangle with length $(2x + 3)$ cm and width $(x - 1)$ cm.
The area of the rectangle is $45\,\text{cm}^2$.



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(a) Show that $2x^2 + x - 48 = 0$.

[2]

(b) Solve the equation $2x^2 + x - 48 = 0$.
You must show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

(c) Make x the subject of the formula.

$$y = \frac{ax + 5}{x - 3}$$

$x =$ ..... **[4]**

22 Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{3}{x-2} + \frac{4}{2x+1} = 2$$

$x =$ ..... , $x =$ ..... **[6]**

23 Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{5}{x-2} - \frac{3}{2x+1} = 2$$

$x =$ ..... , $x =$ ..... **[7]**

24 **(a)** Show that $\dfrac{3}{x-1} - \dfrac{2}{x+3} = 1$ simplifies to $x^2 + x - 14 = 0$.

[4]

(b) Solve the equation $x^2 + x - 14 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

25 **(a)** A train travels 120 km at an average speed of x km/h.
It then travels a further 150 km at an average speed of $(x + 10)$ km/h.
The total time for the journey is 4 hours.
Write down an equation in x and show that it simplifies to $2x^2 - 115x - 600 = 0$.

[4]

(b) Solve the equation $2x^2 - 115x - 600 = 0$ to find the value of x.
Give your answer correct to 1 decimal place.

$x =$ ..... **[2]**

(c) Use your graphic display calculator to solve the equation.

$$2^x - 3 = \frac{5}{x}$$

Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[2]**

26 **(a)** Solve the equation.

Give your answers in the exact form $\frac{p \pm \sqrt{q}}{2}$.

$$\frac{x}{x+2} = \frac{3}{x-6}$$

$x =$ ..... **[5]**

(b) Rearrange the formula to make x the subject.

$$y = \frac{2x-5}{3x+1}$$

$x =$ ..... **[4]**

27 A bus travels 100 km at an average speed of x km/h.
A car travels 100 km at an average speed of $(x + 15)$ km/h.
The car takes 30 minutes less than the bus.

(a) Write down an equation in x and show that it simplifies to $x^2 + 15x - 3000 = 0$.

[3]

(b) Solve the equation $x^2 + 15x - 3000 = 0$.
You must show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

(c) Work out the time taken by the car to travel the 100 km.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

28 **(a)** Make x the subject of the formula.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

(b) Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{5}{x - 2} - \frac{3}{2x + 1} = 2$$

$x =$ ..... , or $x =$ ..... **[6]**

29 Consider the following algebraic equations and formulas.

(a) Rearrange the formula to make x the subject.

$$y = \frac{3x - 1}{x + 2}$$

$x =$ ..... **[4]**

(b) Solve the equation.
Give your answers correct to 2 decimal places.
Show all your working.

$$\frac{2}{x - 1} + \frac{3}{x + 2} = 1$$

$x =$ ..... or $x =$ **[6]**

30 A train travels 150 km at an average speed of x km/h.

(a) Write down an expression, in terms of x, for the time taken in hours.

..... hours **[1]**

(b) A bus travels the same distance of 150 km at an average speed of $(x - 15)$ km/h.
The bus takes 30 minutes longer than the train.
Show that x satisfies the equation $x^2 - 15x - 4500 = 0$.

[4]

(c) Solve the equation $x^2 - 15x - 4500 = 0$ to find the average speed of the train.

..... km/h **[3]**

(d) Rearrange the formula to make q the subject.

$$P = \frac{2q + 3}{q - 1}$$

$q =$ ..... **[4]**

31 Solve the simultaneous equations.

$2x - y = 7$

$3x + 2y = 14$

$x =$ ..... , $y =$ ..... **[3]**

32 Rearrange the formula to make x the subject.

$$y = \frac{2x + 3}{x - 5}$$

$x =$ ..... **[4]**

33 Solve the equation.
Give your answers correct to 2 decimal places.

$$\frac{3}{x+2} + \frac{1}{x-1} = 2$$

$x =$ ..... , $x =$ ..... **[6]**

34 **(a)** Solve the equation.

$$3(2x - 1) = 5x + 8$$

$x =$ ..... **[2]**

(b) Make x the subject of the formula.

$$y = \frac{2x + 3}{5}$$

$x =$ ..... **[3]**

35 **(a)** Solve the simultaneous equations.
You must show all your working.

$2x + 3y = 13$
$5x - 2y = 4$

$x =$ ..... , $y =$ ..... **[3]**

(b) Make x the subject of the formula.

$$y = \frac{3x - 1}{x + 2}$$

$x =$ ..... **[4]**

36 **(a)** Solve the equation.

$$\frac{3}{x-1} + \frac{2}{2x+1} = 2$$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[5]**

(b) The sum of the squares of two consecutive positive even numbers is 340.

Write down an equation in x and solve it to find the two numbers.

The numbers are ..... , and ..... **[4]**

37 **(a)** Solve the equation.

$$3(2x - 5) = 4x + 7$$

$x =$ ..... **[3]**

(b) Solve the simultaneous equations.
You must show all your working.

$$2x + 3y = 13 \qquad 5x - 2y = 4$$

$x =$ ..... , $y =$ ..... **[3]**

38 **(a)** Rearrange the formula to make x the subject.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

(b) Solve the equation.

$$\frac{5}{x - 2} = \frac{2}{x + 1}$$

$x =$ ..... **[3]**

39 **(a)** A car travels 120 km at an average speed of x km/h.
A train travels the same distance at an average speed of $(x + 10)$ km/h.
The time taken by the car is 36 minutes more than the time taken by the train.

(i) Show that $x^2 + 10x - 2000 = 0$.

[3]

(ii) Solve the equation $x^2 + 10x - 2000 = 0$.

$x =$ ..... or $x =$ ..... **[2]**

(b) Solve the equation.

$3x^2 - 7x - 5 = 0$ Give your answers correct to 2 decimal places.
You must show all your working.

$x =$ ..... or $x =$ ..... **[3]**